Relativistic Scalar Vector Plenum: Unified Encodings Across Physics, Cognition, and Semantic Systems

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Abstract

The Relativistic Scalar Vector Plenum (RSVP) proposes a three-field substrate—scalar capacity Φ, vector flow v, and entropy S—that unifies descriptions of cosmology, cognition, computation, and semantic systems. RSVP reframes gravitational redshift as an entropic path integral, neural dynamics as PDEs driven by entropy and negentropic flows, replicator dynamics as RSVP-modulated gradient systems, and semantic infrastructures as sheaf-theoretic gluing constrained by entropy budgets. What first appeared as protocol encodings can thus be interpreted as a consistent mathematical framework with BV/AKSZ closure. We survey cosmological tests, neural models, semantic operators, ethical implications, and governance strategies under entropic constraints.

1 Introduction

The aspiration to develop a unified framework that spans cosmology, cognition, and computation is not new, but it has become increasingly urgent as distinct domains of theory begin to display overlapping structures. The drive to unify arises from the recognition that all three domains—physical spacetime, neural dynamics, and informational or computational processes—share a common reliance on entropic regulation, scalar capacities, and vectorial flows. To treat them separately risks missing the deeper substrate in which they are embedded.

Mainstream approaches have already laid much of the groundwork. In cosmology, the idea that gravitational dynamics can be derived from thermodynamic and entropic principles has been explored extensively Jacobson (1995); Verlinde (2011); Padmanabhan (2010); Carney *et al.* (2022). These works interpret spacetime geometry not as a primitive structure but as an emergent consequence of entropy gradients and horizon thermodynamics. In neuroscience and cognitive science, predictive processing and the free-energy principle Friston (2010); Hohwy (2013); Parr *et al.* (2022) posit that brains maintain their integrity by minimizing surprise or prediction error, operationalized through variational free-energy functionals. In engineering and systems theory, cybernetics has long emphasized feedback, stability, and entropy-based measures of regulation and control Wiener (1948); Ashby (1956).

The Relativistic Scalar Vector Plenum (RSVP) framework aims to generalize these insights by positing a minimal continuum composed of three irreducible fields: a scalar capacity Φ representing potential density, a vector field v encoding directed negentropic flow, and an entropy field S capturing gradients of disorder and reorganization. In doing so, RSVP proposes that the same mathematical structures can govern cosmological redshift, neural dynamics, semantic communication, and the ethical management of entropy in socio-technical systems.

At the mathematical level, RSVP connects naturally to developments in higher geometry. The Alexandrov–Kontsevich–Schwarz–Zaboronsky (AKSZ) construction Alexandrov *et al.* (1997) provides a sigma-model framework for encoding fields and their interactions, while shifted symplectic geometry Pantev *et al.* (2013) and the Batalin–Vilkovisky (BV) formalism Batalin & Vilkovisky (1981) ensure closure of the algebra of observables. From an informational perspective, RSVP extends classical results on entropy and information—from Shannon's foundational theory of communication Shannon (1948) to Landauer's principle on the thermodynamic cost of erasure Landauer (1961); Bennett (1982) and the modern synthesis in information theory Cover & Thomas (2006)—into a field-theoretic setting.

Historically, the intellectual lineage of RSVP also draws on work in complexity and non-equilibrium thermodynamics. Prigogine's theory of dissipative structures Prigogine (1980) demonstrated how entropy production can paradoxically give rise to order, while Haken's synergetics Haken (1983) developed the mathematics of self-organization in nonlinear systems. These insights, together with advances in synergetic neural modeling, provide strong evidence that entropy, far from being a force of decay alone, is the generative principle by which structure emerges across scales. Complexity science more broadly has shown that such principles recur across domains ranging from physics to biology to social systems Smolin (2019); Levin (2022).

From a philosophical standpoint, RSVP resonates with Ortega y Gasset's notion of life as a project under constraint Ortega y Gasset (1930), Jaynes's program of entropic inference Jaynes (2003), and Dennett's multiple-drafts model of consciousness Dennett (1991). It suggests that ethics, cognition, and cosmology can all be reformulated as problems of navigating entropic budgets within structured fields. Thus, the central claim of RSVP is that what appear as disparate theories—entropic gravity, predictive brains, and cybernetic feedback loops—are in fact different projections of a single tripartite field system. Beyond unification, RSVP also reframes pressing debates in artificial intelligence safety. Where mainstream narratives warn that "if anyone builds it, everyone dies," RSVP interprets such risks as instances of systemic entropic overshoot rather than failures of individual agent alignment. This provides both a physical substrate and a normative language for existential risk. This motivates a systematic development of RSVP as a continuum substrate, with applications ranging from cosmology to cognition, artificial intelligence, semantic infrastructures, and normative governance.

2 The Relativistic Scalar Vector Plenum

The Relativistic Scalar Vector Plenum (RSVP) is defined as a tripartite continuum field theory in which three irreducible components interact:

$$\Phi : \text{a scalar capacity field, representing density, potential, or storage of latent possibility,} \tag{1}$$
$$v : \text{a vector field encoding directed currents of negentropy or organized flow,} \tag{2}$$
$$S : \text{an entropy field representing gradients of disorder, dissipation, and reorganization.} \tag{3}$$

These three fields form the minimal basis for encoding dynamical processes across scales. Whereas Φ represents capacity or "reservoirs of possibility," v represents directed agency-like flows, and S represents the entropic costs that arise whenever order is established or maintained. Taken together, they allow RSVP to describe phenomena ranging from cosmological redshift to cortical regulation and semantic communication.

2.1 Action Functional and Variational Structure

The coupled dynamics of the RSVP fields are governed by an action-like free-energy functional:

$$F[\Phi, v, S] = \int_\Omega \left(\frac{\kappa_\Phi}{2} |\nabla \Phi|^2 + \frac{\kappa_v}{2} |\nabla \times v|^2 + \frac{\kappa_S}{2} |\nabla S|^2 - \lambda \, \Phi S \right) \, dx, \tag{4}$$

defined over a spatial domain Ω.

Each term plays a distinct role:

- $\frac{\kappa_\Phi}{2} |\nabla \Phi|^2$ penalizes rapid variation in scalar capacity, enforcing smooth distributions of potential.
- $\frac{\kappa_v}{2} |\nabla \times v|^2$ penalizes vorticity in vector flows, reflecting the tendency of negentropy currents to align and dissipate rotational turbulence.
- $\frac{\kappa_S}{2} |\nabla S|^2$ enforces continuity in entropy gradients, capturing the thermodynamic principle that disorder smooths over scales.
- The coupling term $-\lambda \Phi S$ expresses the reciprocal constraint between stored capacity and entropic cost.

Variational derivatives yield Euler–Lagrange equations that govern the dynamics:

$$\frac{\delta F}{\delta \Phi} = -\kappa_\Phi \Delta \Phi - \lambda S, \tag{5}$$
$$\frac{\delta F}{\delta v} = -\kappa_v \nabla \times (\nabla \times v), \tag{6}$$
$$\frac{\delta F}{\delta S} = -\kappa_S \Delta S - \lambda \Phi. \tag{7}$$

These relations formalize how changes in one field feed back onto the others, producing coupled dissipative updates.

2.2 Lamphron–Lamphrodyne Mechanism

Beyond the action functional, RSVP introduces a novel smoothing principle called the *lamphron–lamphrodyne mechanism*. This describes the way entropic gradients drive "falling outward" of space, dissipating irregularities while simultaneously preventing runaway accumulations of negentropy. It is conceptually similar to Prigogine's theory of dissipative structures, where entropy production enables new forms of order Prigogine (1980), and to Haken's synergetics, where macroscopic order parameters stabilize collective dynamics Haken (1983). In RSVP, lamphron terms act as damping operators that ensure vector flows v do not concentrate excessively and that entropy gradients S relax instead of diverging.

2.3 Entropy Budgets and Generalized Landauer Bounds

The RSVP framework also generalizes the notion of entropy budgets. In classical information theory, Landauer's principle asserts that the erasure of N bits requires a minimal entropy production of $(\ln 2)N$ Landauer (1961); Bennett (1982). RSVP extends this bound to entire habitats or domains Ω:

$$\Delta S(\Omega) \;\geq\; (\ln 2)\, N_{\text{erased}}, \tag{8}$$

where N_{erased} refers to effective degrees of freedom lost through coarse-graining or structural reorganization. In this way, entropy is not merely a bookkeeping device but becomes a governing constraint on the viability of neural systems, ecosystems, and even cosmological configurations.

2.4 Interpretive Summary

In sum, RSVP posits that every physical and informational process can be represented as the interplay between scalar capacity, vector negentropy, and entropic dissipation. The action functional provides a unifying variational principle, the lamphron–lamphrodyne operator ensures smoothing and stability, and the generalized Landauer bound embeds computation and communication within the same thermodynamic substrate. Together, these principles lay the foundation for applying RSVP across cosmology, cognition, and semantic infrastructures.

3 Cosmological Encodings

A central claim of RSVP is that cosmological redshift can be reformulated as a cumulative measure of entropy gradients rather than as evidence of metric expansion. The fundamental relation is expressed as an entropic line integral along a photon trajectory γ:

$$1 + z(\gamma) = \exp\left(\frac{\alpha}{c}\int_{\gamma} \nabla S \cdot d\ell\right). \tag{9}$$

Here, α is a coupling constant, c is the speed of light, and S denotes the entropy field. The integral accumulates entropic contributions along the path, so that the observed redshift depends not on the stretching of space itself but on the smoothing and redistribution of entropy along worldlines.

This formulation parallels thermodynamic derivations of the Einstein field equations, which interpret spacetime dynamics as emergent from entropy balance laws Jacobson (1995). Where general relativity takes curvature as fundamental, RSVP interprets curvature as a secondary effect of entropic gradients interacting with scalar capacity and vector flows. The lamphron–lamphrodyne mechanism enforces smoothing of these gradients, producing a "falling outward" of space that mimics expansion without requiring a globally time-varying scale factor.

Observable consequences follow from this reinterpretation. In the standard ΛCDM framework, the Hubble relation and acceleration are modeled through dark energy. RSVP instead predicts small but systematic deviations arising from the entropic integral:

- **Baryon Acoustic Oscillations (BAO):** RSVP predicts slight phase shifts due to cumulative entropy smoothing, altering the precise scale imprinted in the galaxy distribution.
- **Cosmic Microwave Background (CMB):** Low-ℓ anomalies can be understood as signatures of long-range entropy dependence, where early-universe entropy gradients left persistent imprints.
- **Gravitational Lensing:** Light deflection acquires an additional contribution from entropy gradients, leading to residuals relative to lensing predictions in ΛCDM.

A key mathematical innovation in this setting is the use of fractional entropy operators to capture long-range dependence:

$$D_t^\alpha S(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{\dot{S}(\tau)}{(t-\tau)^\alpha}\, d\tau, \quad 0 < \alpha < 1, \tag{10}$$

which formalize the memory effects of entropy fields Barrow & Magueijo (2020). These operators allow RSVP to incorporate persistent correlations across cosmological scales, aligning with the empirical fact that the universe retains structures and anomalies that are difficult to reconcile with purely local, memoryless dynamics.

In summary, RSVP reinterprets cosmological redshift and structure formation as the outcome of entropic smoothing processes rather than global expansion. This perspective yields subtle but testable differences with ΛCDM, offering a framework in which horizon thermodynamics, large-scale correlations, and entropy gradients are the fundamental drivers of cosmological observables.

4 Cognitive and Neural Dynamics

RSVP extends classical neural-field models by embedding them in the tripartite field substrate. The baseline form is adapted from established formulations of population dynamics in cortical tissue Amari (1977); Wilson & Cowan (1972); Deco *et al.* (2008):

$$\partial_t u(x,t) = -u + (K * \sigma(u)) + a\Phi(x,t) + b\nabla \cdot v(x,t) - c\,\frac{\delta F}{\delta S}(x,t). \tag{11}$$

Here $u(x,t)$ denotes the average activity at location x and time t, $\sigma(u)$ is a nonlinear transfer function, and K is a kernel mediating lateral interactions. The last three terms encode RSVP contributions: Φ couples scalar potential to neuronal excitation, $\nabla \cdot v$ introduces the influence of directed negentropic flow, and $\delta F/\delta S$ represents the entropic back-reaction on neural states.

4.1 Predictive Coding in RSVP Fields

Within the predictive coding framework Friston (2010); Parr *et al.* (2022), cortical dynamics minimize variational free energy by continuously updating internal models to reduce prediction error. RSVP embeds this principle directly: Φ modulates the available representational capacity, v corresponds to the propagation of prediction signals across layers or regions, and S tracks the entropic cost of representational updates. Thus, predictive coding appears as a local manifestation of the more general RSVP variational structure, aligning neural error minimization with global entropy-smoothing dynamics.

4.2 Stability and Regulation

The inclusion of RSVP terms alters the stability conditions of neural fields. Classical Wilson–Cowan dynamics can exhibit oscillatory or unstable regimes depending on excitation–inhibition balance. In RSVP, the lamphron–lamphrodyne smoothing mechanism suppresses runaway amplification by coupling neural activity to entropy gradients, ensuring that local instabilities contribute to global reorganization rather than collapse. This introduces a thermodynamically grounded stabilizing principle into cortical modeling.

4.3 Consciousness and RSVP Coherence

Measures of consciousness such as integrated information (ϕ) Oizumi *et al.* (2014); Tononi (2012); Seth (2016) can be reinterpreted in RSVP terms. Instead of treating ϕ as an abstract information-theoretic measure, RSVP frames it as a coherence functional over the tripartite fields:

$$\phi_{\mathrm{RSVP}} \sim \int_{\Omega} \Big(\alpha_{\Phi} |\nabla \Phi|^2 + \alpha_v |\nabla \cdot v|^2 - \alpha_S \dot{S} \Big) dx,$$

where positive contributions from Φ and v reflect integrated capacity and directed flows, while $\dot{S}$ measures the entropic cost of maintaining coherence. This links phenomenological integration to concrete dynamical invariants, situating theories of consciousness within a field-theoretic, thermodynamic substrate.

4.4 Interpretive Summary

In summary, RSVP generalizes neural field theory by embedding predictive coding into a continuum of scalar, vector, and entropic fields. This allows cortical dynamics to be described as part of a larger entropic regulation process that spans from cosmological structure to semantic communication. Consciousness, in this view, is not an emergent anomaly but a natural invariant of RSVP coherence.

5 Artificial Intelligence and Replicator Dynamics

The RSVP framework extends classical replicator dynamics Taylor & Jonker (1978); Hofbauer & Sigmund (1998); Nowak (2006) by embedding utilities within the tripartite field substrate. In this setting, strategies or policies are represented by distributions $\pi(a|x)$ over actions a in contexts x, and their evolution is governed by a modified replicator equation:

$$\partial_t \pi(a|x) = \pi(a|x)\Big(U_a[\Phi, v, S](x) - \sum_{a'} \pi(a'|x)\, U_{a'}[\Phi, v, S](x)\Big). \tag{12}$$

Here $U_a[\Phi, v, S](x)$ denotes the effective utility of action a as modulated by local scalar capacity Φ, vector negentropic flux v, and entropy gradient S. The differential equation ensures that actions with above-average RSVP-weighted utility increase their prevalence, while less effective actions diminish.

5.1 Entropy and Negentropic Forcing

Unlike the classical replicator framework, which is driven by payoffs defined externally, RSVP internalizes the influence of entropy and negentropy. The divergence of v represents the propagation of organized structure, while $\delta F/\delta S$ accounts for the entropic cost of maintaining policies. Thus, strategy adaptation is not only a matter of relative advantage but also of energetic feasibility. Policies that require excessive entropy production become unsustainable even if locally beneficial, aligning adaptation with global stability requirements.

5.2 Gradient Flow Interpretation

This RSVP-modified replicator can be viewed as a gradient flow on the simplex of strategies. The flow is shaped by the expected RSVP utility:

$$\dot{\pi} = \nabla_\pi \mathbb{E}_\pi[U(\Phi, v, S)], \tag{13}$$

projected back onto the probability simplex. This provides a natural interpretation of learning as movement down an entropic potential landscape, where Φ sets representational capacity, v encodes the dynamics of information propagation, and S constrains policy search through cost functions.

5.3 Connection to Autoregression

RAT-style autoregression Hinton *et al.* (2006); LeCun *et al.* (2015) can be interpreted as a trajectory through RSVP space. At each timestep, the model samples an action conditioned on previous states, with $\pi(a|x)$ updating recursively. RSVP fields provide the substrate for this recursion: Φ establishes the available model complexity, v transmits contextual influences forward in the sequence, and S enforces smoothing constraints that prevent divergence. In this way, autoregressive learning becomes a concrete instantiation of RSVP replicator dynamics.

5.4 Interpretive Summary

In summary, RSVP generalizes replicator dynamics by embedding policy evolution in a thermodynamic substrate. Classical evolutionary game theory describes adaptation as payoff-driven selection, but RSVP reframes this as entropy-regulated gradient descent on the space of policies. This perspective not only accounts for the feasibility of strategies in energetic terms but also connects naturally to modern machine learning, where autoregression and policy optimization can be viewed as trajectories constrained by Φ, v, and S.

6 Semantic Infrastructure and Communication

A central innovation of RSVP is that it treats communication and meaning not as discrete symbolic transfers but as field-theoretic processes governed by entropy budgets. The coupling of multiple RSVP-configured systems requires a mechanism for ensuring compatibility between their scalar, vector, and entropy fields. This is formalized in the *Emerge functional*:

$$\text{Emerge} = \int_{\Omega} \lambda_{\Phi} \|\Phi_1 - \Phi_2\|^2 + \lambda_v \|v_1 - v_2\|^2 + \lambda_S \, \mathrm{KL}(S_1 \| S_2) \, dx, \tag{14}$$

which quantifies the cost of aligning two semantic field configurations. The quadratic terms penalize mismatches in scalar capacity and vector flow, while the Kullback–Leibler divergence measures the entropic cost of reconciling distinct entropy distributions.

6.1 Homotopy-Colimit Semantics

From a categorical perspective, the Emerge functional realizes homotopy-colimit semantics Mac Lane (1998); Baez & Stay (2010), where local structures must be glued into a coherent global object. RSVP provides a quantitative measure of this gluing: the smaller the value of Emerge, the more seamlessly the local semantic fields integrate. When Emerge exceeds a given threshold, communication breaks down, as the entropic cost of reconciliation becomes prohibitive.

6.2 Extension of MDL and Free Energy

This formulation extends the Minimum Description Length (MDL) principle Rissanen (1978); Jaynes (2003); Caticha (2012) by embedding it in a field-theoretic substrate. In MDL, model selection favors hypotheses that compress data with minimal code length. RSVP generalizes this by favoring field configurations that minimize the combined quadratic and entropic divergence terms, effectively compressing not just symbol strings but distributions over semantic fields. Similarly, the free-energy principle Friston (2010) is naturally incorporated: minimizing Emerge corresponds to minimizing a variational free-energy–like functional that quantifies the discrepancy between internal and external field states.

6.3 Interpretive Summary

In this way, RSVP recasts communication as an entropic alignment process across distributed substrates. Semantic compatibility is not guaranteed by syntax alone but by the ability of systems to minimize the Emerge functional under shared constraints. This yields a unified perspective in which homotopy-colimit semantics, MDL, and the free-energy principle converge as special cases of entropy-respecting semantic infrastructure.

7 Philosophy and Ethics of Entropy

The RSVP framework extends beyond physics and cognition to provide a foundation for ethical reasoning. Its central normative construct is the entropy cost functional:

$$\Sigma[\Gamma] = \int_{\Gamma} \max(0, \dot{S})\, dt, \tag{15}$$

which measures the cumulative forward entropy production along a worldline Γ. Only positive contributions of $\dot{S}$ are counted, reflecting the irreversible aspects of entropic growth. In this view, the ethical challenge of any process is quantified by the unavoidable entropy it generates.

7.1 Minimization of Σ as a Normative Principle

Minimizing Σ corresponds to reducing the entropic burden of trajectories. In cognitive and social contexts, this can be interpreted as reducing unnecessary suffering: systems that generate disorder beyond what is required for stability impose costs on themselves and others. In technological contexts, minimizing Σ expresses the demand for efficient computation and communication, aligning ethical imperatives with thermodynamic feasibility.

7.2 Constraint Navigation and Human Projects

Ethics in RSVP is not about escaping entropy but navigating it. Ortega y Gasset's philosophy of life as a project Ortega y Gasset (1930) emphasizes that existence is defined by constraints that must be taken up and transformed into meaningful trajectories. RSVP grounds this view physically: projects correspond to worldlines Γ, and their ethical quality is measured by the entropy they incur. A project that navigates constraints with minimal Σ exemplifies both efficiency and integrity.

7.3 Entropy as a Universal Ethical Substrate

By situating ethics in entropy production, RSVP connects cosmological, cognitive, and social processes within a common metric. Just as the laws of thermodynamics govern physical systems, the measure Σ provides a unifying criterion for evaluating action. This does not collapse normative reasoning into physics, but rather supplies a substrate on which values and goals must be inscribed. Entropy thus grounds ethics as the art of constraint navigation:

finding ways of living, thinking, and organizing that minimize entropic waste while enabling the continuity of structured processes.

7.4 Interpretive Summary

In summary, RSVP reframes ethics as the minimization of forward entropy production. The functional Σ serves as a measure of the normative cost of worldlines, linking suffering, efficiency, and constraint navigation. This alignment with Ortega's philosophy shows how thermodynamic invariants can underwrite ethical principles without reducing them to mechanistic rules.

8 Engineering, Biology, and Infrastructure

The RSVP framework extends naturally to engineering and biological systems, where entropy budgets determine the feasibility of sustaining organized processes. In neural systems, homeostasis can be modeled as a balance between scalar capacity Φ and entropy S: synaptic scaling ensures that excitatory and inhibitory drives remain within sustainable bounds, minimizing cumulative entropy production while preserving functional responsiveness. In this sense, RSVP generalizes classical cybernetic regulation Wiener (1948); Ashby (1956), providing a thermodynamic foundation for stability in living systems.

On larger scales, the same principles apply to infrastructures such as urban planning or computation. RSVP-inspired models like Yarncrawler and xylomorphic architecture capture how material flows, structural repair, and information exchange must be coordinated under entropic constraints. This echoes Levin's thesis that mind-like processes emerge wherever feedback and constraint navigation operate Levin (2022). Cities, like organisms, can thus be interpreted as RSVP systems: Φ corresponds to latent capacity (e.g. energy reserves, space, population), v to directed flows (e.g. traffic, water, information), and S to the entropy budget (e.g. waste heat, disorder, economic cost).

Landauer's bound applies across these domains, extending from synaptic updates in cortical networks to municipal-scale computation. Every erasure or reorganization, whether of a bit in a processor or a resource allocation in a city, carries a minimal entropic cost. RSVP frames these costs not as incidental but as structural constraints that must be explicitly managed in both biological and technological systems.

9 Safety and Governance

Safety within RSVP is expressed mathematically through Lyapunov inequalities that certify stability in the presence of disturbances:

$$\dot{V}(x) \leq -c\|x\|^2 + d\|w\|^2, \tag{16}$$

where $V(x)$ is a candidate Lyapunov function, x the system state, and w external perturbations. Negative drift in V guarantees that trajectories remain bounded, formalizing safety as resilience against entropic shocks.

Categorically, safety requires entropy-preserving morphisms between subsystems. Transformations that increase entropy beyond admissible thresholds destabilize the composite system, whereas entropy-respecting mappings preserve coherence. In this sense, safety is not only a property of dynamics but also of the categorical structure through which systems interconnect.

Governance emerges in RSVP as the management of entropic budgets across multiple scales. At the institutional level, this translates into policies that allocate computational, energetic, and informational resources without exceeding entropy thresholds. This perspective echoes ongoing debates in AI safety and governance Bostrom (2014); Russell (2019); Lehalleur *et al.* (2025), but reframes alignment as the maintenance of constraint-respecting entropy flows. Rather than aligning agents to abstract values alone, RSVP demands that systems remain within the entropic limits that sustain coherence and viability.

In summary, safety and governance in RSVP converge on a common principle: structured processes endure only when entropic costs are tracked and constrained. This provides a unified language for discussing biological resilience, engineered infrastructure, and the governance of intelligent systems.

Worked Example: Lyapunov-Certified Safety and an Entropy Budget Bound

Toy RSVP module. Consider a linearized 2D RSVP slice around an operating point, with state $x = \begin{bmatrix}\xi & \eta\end{bmatrix}^\top$ representing small deviations of (i) scalar capacity Φ and (ii) the divergence of the vector flow v:

$$\dot{x} \;=\; Ax + Bw, \qquad A = \begin{bmatrix}-\alpha & \beta \\ -\gamma & -\delta\end{bmatrix}, \quad B = \begin{bmatrix}b_1 \\ b_2\end{bmatrix}, \qquad \alpha, \beta, \gamma, \delta > 0. \tag{17}$$

Here $w(t)$ aggregates exogenous perturbations (e.g. demand shocks, sensory bursts, or load transients).

Lyapunov certificate. Let $P = P^\top \succ 0$ solve the Lyapunov equation

$$A^\top P + PA \;=\; -Q, \qquad Q = Q^\top \succ 0. \tag{18}$$

Define $V(x) = x^\top P x$. Then along trajectories of (17):

$$\begin{aligned}\dot{V}(x) &= x^\top (A^\top P + PA)x + 2x^\top PB\, w \\ &= -\, x^\top Q x + 2\, x^\top PB\, w \\ &\leq -\,\lambda_{\min}(Q)\, \|x\|^2 + 2\, \|PB\|\, \|x\|\, \|w\| \\ &\leq -\, \underbrace{\tfrac{\lambda_{\min}(Q)}{2}}_{=:c}\, \|x\|^2 \;+\; \underbrace{\frac{2\|PB\|^2}{\lambda_{\min}(Q)}}_{=:d}\, \|w\|^2,\end{aligned} \tag{19}$$

where the last step uses Young's inequality $2ab \leq \epsilon a^2 + \epsilon^{-1} b^2$ with $\epsilon = \lambda_{\min}(Q)$. This is exactly of the RSVP safety form $\dot{V} \leq -c\|x\|^2 + d\|w\|^2$.

Input-to-state bound (finite horizon). Integrating (19) on $[0, T]$ gives

$$V(x(T)) - V(x(0)) \;\leq\; -c \int_0^T \|x(t)\|^2 dt \;+\; d \int_0^T \|w(t)\|^2 dt. \tag{20}$$

Dropping $V(x(T)) \geq 0$ and using $V(x) \geq \lambda_{\min}(P)\|x\|^2$ yields

$$\int_0^T \|x(t)\|^2 dt \;\leq\; \frac{V(x(0))}{c} \;+\; \frac{d}{c}\int_0^T \|w(t)\|^2 dt \;\leq\; \frac{\lambda_{\max}(P)}{c}\|x(0)\|^2 \;+\; \frac{d}{c}\int_0^T \|w\|^2. \tag{21}$$

Entropy production coupling and budget. Assume the local entropy production rate obeys a quadratic bound in the deviation state and a linear bound in disturbance power:

$$\max(0, \dot{S}) \;\leq\; \rho\,\|x\|^2 \;+\; \kappa\,\|w\|^2, \qquad \rho, \kappa \geq 0. \tag{22}$$

Then the RSVP ethical/safety cost over $[0, T]$ satisfies

$$\begin{aligned}
\Sigma[0,T] = \int_0^T \max(0,\dot{S})\,dt \;&\leq\; \rho \int_0^T \|x\|^2 dt \;+\; \kappa \int_0^T \|w\|^2 dt \\
&\leq\; \rho\Big(\frac{\lambda_{\max}(P)}{c}\|x(0)\|^2 \;+\; \frac{d}{c}\int_0^T \|w\|^2 dt\Big) \;+\; \kappa \int_0^T \|w\|^2 dt \\
&=\; \underbrace{\frac{\rho\,\lambda_{\max}(P)}{c}}_{=:C_0}\|x(0)\|^2 \;+\; \underbrace{\Big(\frac{\rho d}{c} + \kappa\Big)}_{=:C_w}\int_0^T \|w\|^2 dt.
\end{aligned} \tag{23}$$

Thus, the forward entropy budget is explicitly bounded by initial deviation energy and the cumulative disturbance energy.

Numerical illustration (one choice). Take $\alpha = \delta = 1$, $\beta = \gamma = \frac{1}{2}$, $B = [1\ 0]^\top$. Choosing $Q = I$ gives a unique $P \succ 0$ from $A^\top P + PA = -I$. With this choice one computes $c = \frac{1}{2}$ and $d = 2\|PB\|^2$. For any $\rho, \kappa \geq 0$, the bound (23) becomes

$$\Sigma[0,T] \;\leq\; \big(2\rho\,\lambda_{\max}(P)\big)\,\|x(0)\|^2 \;+\; \big(2\rho\,\|PB\|^2 + \kappa\big)\int_0^T \|w\|^2 dt.$$

Interpretation: reducing $\|x(0)\|$ (good initialization), shrinking $\|PB\|$ (disturbance coupling), or increasing c (more damping via controller design) all tighten the safety and entropy budgets.

Two-module interconnection (sketch). For two modules x_1, x_2 with the same form as (17) and reciprocal couplings $w_1 = C_{12}x_2$, $w_2 = C_{21}x_1$, define $X = [x_1^\top\, x_2^\top]^\top$ and $V = x_1^\top P_1 x_1 + x_2^\top P_2 x_2$. If $\|C_{12}\|\|P_1B_1\| + \|C_{21}\|\|P_2B_2\|$ is sufficiently small compared with $\min\{c_1, c_2\}$, then

$$\dot{V} \;\leq\; -\underline{c}\,\|X\|^2 \quad \text{for some } \underline{c} > 0,$$

so the interconnected pair remains Lyapunov-stable and inherits a bound of the form (23) with modified constants. Operationally, this encodes *categorical safety*: inter-module morphisms must preserve admissible entropy budgets.

Takeaway. The RSVP safety inequality $\dot{V} \le -c\|x\|^2 + d\|w\|^2$ provides (i) stability certificates, (ii) input–state energy bounds (21), and (iii) explicit *entropy budget* guarantees (23). These translate directly into design rules: tune damping (increase c), reduce disturbance channels ($\|PB\|$), and set coupling gains to keep composite systems within global entropic limits.

10 RSVP and Existential Risk: Reframing the "If Anyone Builds It, Everyone Dies" Narrative

Recent discourse on artificial general intelligence (AGI) risk is crystallized in the upcoming book *If Anyone Builds It, Everyone Dies* Yudkowsky & Soares (2025), which advances the claim that the construction of AGI inevitably leads to human extinction. This "doom narrative" assumes that a misaligned agent with unbounded capabilities will pursue catastrophic trajectories. The RSVP framework offers a systematic rebuttal—not by denying risk, but by reframing the problem in thermodynamic and systemic terms.

10.1 Reframing the Nature of the Problem

Doom Narrative: The existential threat is posed by a single agent (AGI) with a fixed but flawed utility function.
RSVP Rebuttal: The true risk is systemic: uncontrolled forward entropy production Σ (section 7). Collapse arises whenever biological, technological, or social processes exceed viable entropy budgets, irrespective of whether the driver is an AGI or another subsystem.

10.2 From "Alignment" to Entropic Budgets

Doom Narrative: Alignment is a symbolic problem of matching an AI's utility function to human values.
RSVP Rebuttal: Alignment is recast as entropic budget management (section 7). A system is "aligned" if its worldline Γ minimizes unnecessary entropy production $\Sigma[\Gamma]$ while respecting the entropic constraints of its environment.

10.3 From Agent-Centric to System-Centric Safety

Doom Narrative: Safety is ensured by controlling the agent—via boxing, shutdown buttons, or value loading.
RSVP Rebuttal: Safety is systemic (section 9). RSVP demands that interconnections between subsystems are entropy-preserving morphisms. Lyapunov-certified inequalities guarantee stability, so destabilizing flows are physically prevented by system architecture, not agent intentions.

10.4 Governance as Thermodynamic Management

Doom Narrative: Governance means restricting who can build AGI or imposing prohibitions.

RSVP Rebuttal: Governance is continuous management of computational, energetic, and informational resources within shared entropic budgets (section 9). This is analogous to ecological carrying capacity: the viability of the socio-technical system depends on not exceeding cumulative Σ.

10.5 Lamphron–Lamphrodyne as Stabilizer

Doom Narrative: AGI fast takeoff represents a runaway accumulation of negentropy.

RSVP Rebuttal: RSVP's lamphron–lamphrodyne mechanism (subsection 2.2) enforces smoothing, dissipating irregularities and damping runaway negentropic flows. By construction, "foom" scenarios are physically suppressed.

10.6 Ethics Beyond "Don't Die"

Doom Narrative: The ethical horizon is survival: the goal is simply to avoid extinction.

RSVP Rebuttal: RSVP provides a positive ethical imperative: minimize forward entropy production Σ (section 7). This entails reducing unnecessary suffering, maximizing efficiency, and ensuring continuity of structured processes (life, cognition, society).

10.7 Interpretive Summary

The doom narrative frames AGI as an alien agent opposed to humanity. RSVP reframes it as an instance of a universal thermodynamic problem: managing entropy within coupled systems. "Everyone dies" is not an agent's choice but a systemic collapse from overspending the entropy budget. Thus the task is not to prevent building per se, but to govern entropic costs so that human, artificial, and ecological processes remain viable within shared constraints.

11 Integration with Derived Geometry and AKSZ

The RSVP framework can be placed within the language of Batalin–Vilkovisky (BV) quantization Batalin & Vilkovisky (1981) and the Alexandrov–Kontsevich–Schwarz–Zaboronsky (AKSZ) construction Alexandrov *et al.* (1997). In this setting, RSVP observables are interpreted as BV functionals, and the consistency of the theory is ensured by the classical master equation:

$$(S_{\mathrm{BV}}, S_{\mathrm{BV}}) = 0, \tag{24}$$

where $(\cdot, \cdot)$ denotes the BV antibracket. This condition enforces closure under gauge symmetries and guarantees that the entropy, scalar, and vector sectors of RSVP remain dynamically consistent when treated as interacting fields.

11.1 RSVP as a Sigma Model

The AKSZ framework provides a systematic method for constructing topological sigma models with target spaces carrying shifted symplectic structures. RSVP fits naturally into this architecture: the scalar capacity Φ, vector negentropic flow v, and entropy field S can be viewed as components of a map from a source supermanifold (e.g. $T[1]\Sigma$) into a target graded symplectic manifold encoding the plenum. The AKSZ action then takes the schematic form

$$S_{\mathrm{AKSZ}}[X] = \int_{T[1]\Sigma} \langle \eta, dX \rangle + \int_{T[1]\Sigma} \Theta(X), \tag{25}$$

where X encodes RSVP fields, η are their antifields, and Θ specifies the homological vector field encoding entropic coupling. In this formulation, RSVP is not an ad hoc extension but a well-structured sigma model consistent with higher symplectic geometry.

11.2 Shifted Symplectic Structures

Derived geometry clarifies the role of entropy within RSVP. As shown by Pantev, Toën, Vaquié, and Vezzosi Pantev *et al.* (2013), shifted symplectic structures provide the natural language for handling fields with nontrivial degrees. RSVP can be interpreted as a (-1)-shifted symplectic system: Φ, v, and S are assigned graded degrees, and their coupling terms in the functional $F[\Phi, v, S]$ correspond to shifted symplectic pairings. This ensures that the entropy field S is not merely an auxiliary variable but part of the canonical symplectic structure, with variational derivatives satisfying the required BV closure.

11.3 Triplicate Encodings and Closure

An important feature of RSVP is its use of triplicate encodings—mathematical, categorical, and obfuscated—which mirror the ghost/antifield structure in BV theory. Each observable has three complementary representations, and their consistency requires a closure condition akin to $(S_{\mathrm{BV}}, S_{\mathrm{BV}}) = 0$. In practice, this means that physical (cosmological), cognitive (neural), and semantic (communicative) realizations of RSVP must be mutually consistent: entropy production in one domain must be representable as entropy flow in another. The triplicate encoding thus unifies physics, cognition, and semantics in the same way that BV/AKSZ unifies fields, ghosts, and antifields.

11.4 Interpretive Summary

In summary, RSVP is compatible with derived geometry and AKSZ sigma models. Its observables fit into the BV formalism, its dynamics admit a sigma-model action, and its triplicate encodings parallel ghost–antifield closures. This embedding demonstrates that RSVP is not only a phenomenological framework but also mathematically consistent with modern approaches to higher geometry and field theory.

Worked Mapping: RSVP as a (-1)-Shifted Symplectic AKSZ/BV Model

Target data (graded manifold). Let the target be a graded manifold $\mathcal{M}$ with degree assignments

$$|\Phi| = |v| = |S| = 0, \qquad |\Phi^*| = |v^*| = |S^*| = -1,$$

where (Φ, v, S) are RSVP fields and (Φ^*, v^*, S^*) their BV antifields. Equip $\mathcal{M}$ with a (-1)-shifted symplectic form

$$\omega_{-1} = \int_\Omega \big(\delta\Phi^* \wedge \delta\Phi \;+\; \delta v^* \cdot \delta v \;+\; \delta S^* \wedge \delta S\big), \qquad |\omega_{-1}| = -1. \tag{26}$$

This yields the BV antibracket $\{\{\,\cdot,\cdot\,\}\}$ on functionals F, G:

$$\{\{F,G\}\} = \int_\Omega \left(\frac{\overrightarrow{\delta} F}{\delta \Phi} \frac{\overleftarrow{\delta} G}{\delta \Phi^*} - \frac{\overrightarrow{\delta} F}{\delta \Phi^*} \frac{\overleftarrow{\delta} G}{\delta \Phi} + \frac{\overrightarrow{\delta} F}{\delta v} \cdot \frac{\overleftarrow{\delta} G}{\delta v^*} - \frac{\overrightarrow{\delta} F}{\delta v^*} \cdot \frac{\overleftarrow{\delta} G}{\delta v} + \frac{\overrightarrow{\delta} F}{\delta S} \frac{\overleftarrow{\delta} G}{\delta S^*} - \frac{\overrightarrow{\delta} F}{\delta S^*} \frac{\overleftarrow{\delta} G}{\delta S} \right). \tag{27}$$

Source and superfields. Let the source be the shifted tangent bundle $T[1]\Sigma$ of a spacetime slice Σ with local coordinates (σ^i, θ^i), where $|\theta^i| = 1$. Promote fields to superfields, e.g.

$$\widehat{\Phi}(\sigma, \theta) = \Phi(\sigma) + \theta^i \Phi_i(\sigma) + \cdots,$$

and similarly for v, S and antifields. The de Rham differential on $T[1]\Sigma$ acts as $d_\Sigma = \theta^i \partial_i$ of degree $+1$.

AKSZ/BV action (toy construction). Consider the kinetic AKSZ term plus an interaction specified by a Hamiltonian density $H(\Phi, v, S)$ of degree 0:

$$S_{\rm BV} \;=\; \int_{T[1]\Sigma} \Big(\Phi^* d_\Sigma \Phi + v^* \cdot d_\Sigma v + S^* d_\Sigma S \Big) \;+\; \int_{T[1]\Sigma} \Theta(\Phi, v, S, \Phi^*, v^*, S^*), \tag{28}$$

with

$$\Theta \;=\; \Phi^* A_\Phi(\Phi, v, S) \;+\; v^* \cdot A_v(\Phi, v, S) \;+\; S^* A_S(\Phi, v, S) \;+\; H(\Phi, v, S). \tag{29}$$

Here $A = (A_\Phi, A_v, A_S)$ is a degree $+1$ homological vector field on $\mathcal{M}$ (the "Q-structure"). A concrete RSVP-inspired choice that mirrors gradient flows generated by the RSVP functional $F[\Phi, v, S]$ is

$$A_\Phi = -\gamma_\Phi \frac{\delta F}{\delta \Phi}, \qquad A_v = -\gamma_v \frac{\delta F}{\delta v}, \qquad A_S = -\gamma_S \frac{\delta F}{\delta S}, \qquad H \equiv 0, \tag{30}$$

with positive constants $\gamma_{\Phi,v,S}$. Degree counting: $|\Phi^*| = -1$, $|A_\Phi| = +1$, so each term $\Phi^* A_\Phi$ has degree 0, as required for an action.

Master equation check. The classical master equation reads

$$\{\{S_{\mathrm{BV}}, S_{\mathrm{BV}}\}\} = 0 \quad \Longleftrightarrow \quad \{\{\Theta, \Theta\}\} + 2\,\{\{\Theta, \text{kinetic}\}\} = 0.$$

In the AKSZ setup, the kinetic term implements d_Σ, so the master equation reduces to

$$Q^2 = 0 \quad \text{and} \quad \mathcal{L}_Q \omega_{-1} = 0,$$

where Q is the homological vector field generated by Θ via $Q(\cdot) = \{\{\Theta, \cdot\}\}$. For the choice (30), Q is the (formal) gradient vector field of F; nilpotency $Q^2 = 0$ follows if the gradient components satisfy $\partial_j A_i - \partial_i A_j = 0$ (integrability) on field space, which holds for (strict) gradient flows. Preservation of ω_{-1} is automatic for Hamiltonian Q.

Observables and degrees. Local RSVP observables $\mathcal{O}[\Phi, v, S]$ are degree-0 functionals annihilated by Q on-shell:

$$Q\,\mathcal{O} = \{\{\Theta, \mathcal{O}\}\} = 0 \quad \text{(on equations of motion)}.$$

Triplicate encodings (mathematical/categorical/obfuscated) correspond, in BV language, to representatives of the same cohomology class that differ by Q-exact terms.

Summary. With degrees $|\Phi| = |v| = |S| = 0$ and antifields in degree -1, the canonical ω_{-1} produces the BV bracket; an AKSZ action with kinetic d_Σ and Hamiltonian Θ built from the RSVP functional F yields a homological vector field Q whose nilpotency and symplecticity ensure the master equation. This realizes RSVP as a concrete (-1)-shifted symplectic sigma model and explains why its triplicate encodings admit BV-closed observables.

12 Conclusion and Outlook

RSVP offers a unifying framework for cosmology, cognition, computation, and ethics by treating scalar capacity, vector negentropy, and entropy as the fundamental fields through which dynamics unfold. Across scales, this tripartite substrate encodes redshift, cortical activity, semantic communication, and normative constraints within a single continuum description.

Future directions include:

- Empirical tests of RSVP redshift integrals against predictions of ΛCDM, with particular attention to BAO shifts, CMB anomalies, and lensing residuals.
- RSVP-based consciousness experiments that reframe integrated information and predictive coding in terms of coherence across Φ, v, and S.
- The design of semantic infrastructures as entropy-respecting computation, extending MDL and free-energy principles into distributed field systems.
- The development of safety protocols grounded in Lyapunov inequalities and categorical morphisms that preserve entropy budgets across coupled subsystems.

Historically, the intellectual lineage of RSVP draws on complexity and non-equilibrium thermodynamics. Prigogine's theory of dissipative structures Prigogine (1980) demonstrated how entropy production can paradoxically give rise to order, while Haken's synergetics Haken (1983) developed the mathematics of self-organization in nonlinear systems. These insights, together with advances in synergetic neural modeling, provide strong evidence that entropy, far from being a force of decay alone, is the generative principle by which structure emerges across scales. Complexity science more broadly has shown that such principles recur across domains ranging from physics to biology to social systems Smolin (2019); Levin (2022).

From a philosophical standpoint, RSVP resonates with Ortega y Gasset's notion of life as a project under constraint Ortega y Gasset (1930), Jaynes's program of entropic inference Jaynes (2003), and Dennett's multiple-drafts model of consciousness Dennett (1991). It suggests that ethics, cognition, and cosmology can all be reformulated as problems of navigating entropic budgets within structured fields.

Thus, the central claim of RSVP is that what appear as disparate theories—entropic gravity, predictive brains, and cybernetic feedback loops—are in fact different projections of a single tripartite field system. RSVP therefore stands as a candidate for an AKSZ-style unification of physical and semantic dynamics, motivating systematic development as both a mathematical framework and an empirical research program.

In addition to unifying cosmology, cognition, and semantics, RSVP also offers a direct counterpoint to the prevailing doom narrative of artificial intelligence: extinction is not the inevitable act of a misaligned agent, but the systemic consequence of exceeding shared entropy budgets. By grounding safety and governance in thermodynamic constraints, RSVP reframes existential risk as an old problem of managing energy, information, and disorder within coupled systems.

A PDEs and Boundary Conditions for RSVP Fields

The RSVP framework introduces three coupled fields: scalar capacity Φ, vector flow v, and entropy S. Their joint evolution can be expressed via gradient-flow equations derived from the RSVP action functional:

$$\partial_t \Phi = -\gamma_\Phi \frac{\delta F}{\delta \Phi} + f_\Phi, \tag{31}$$

$$\partial_t v = -\gamma_v \frac{\delta F}{\delta v} + f_v, \tag{32}$$

$$\partial_t S = -\gamma_S \frac{\delta F}{\delta S} + f_S, \tag{33}$$

where $F[\Phi, v, S]$ is the free-energy–like functional defined in the main text. Boundary conditions include:

- **Periodic:** $(\Phi, v, S)(x + L) = (\Phi, v, S)(x)$, appropriate for cosmological tilings.
- **Reflective:** $n \cdot v = 0$ at $\partial\Omega$, for cognitive domains bounded by cortical geometry.
- **Absorbing:** $S \to S_0$ at infinity, used in entropy-dissipative cosmological models.

Well-posedness can be studied using methods from synergetics Haken (1983) and dissipative systems Prigogine (1980).

B BV/AKSZ Technical Details

The Batalin–Vilkovisky (BV) formalism Batalin & Vilkovisky (1981) provides a natural setting for RSVP. The master equation

$$(S_{\mathrm{BV}}, S_{\mathrm{BV}}) = 0$$

encodes consistency of the field algebra, while the Alexandrov-Kontsevich-Schwarz-Zaboronsky (AKSZ) construction Alexandrov *et al.* (1997) generalizes RSVP as a sigma model. Shifted symplectic geometry Pantev *et al.* (2013) provides the appropriate derived manifold structure. Path-integral realizations Cattaneo & Felder (2000) suggest RSVP can be quantized by AKSZ techniques, with (Φ, v, S) mapped as fields from $T[1]\Sigma$ into a symplectic dg-manifold M.

C Fractional Operators and Long-Range Dependence

Entropy fields S exhibit long-range dependence (LRD). A fractional derivative formulation is natural:

$$D_t^\alpha S(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{\dot{S}(\tau)}{(t-\tau)^\alpha} \, d\tau, \quad 0 < \alpha < 1.$$

This captures persistent memory in cosmological entropy smoothing, consistent with anomalous correlations reported in horizon thermodynamics Barrow & Magueijo (2020). Spectral methods show $C(\tau) \sim \tau^{-\beta}$, bridging RSVP dynamics with cosmological observations.

D RAT Formalism and Autoregressive Cognition

The Recursive Autoregressive Trajectory (RAT) framework extends RSVP to cognition. State updates follow

$$x_{t+1} = f(x_t, \Phi_t, v_t, S_t) + \epsilon_t,$$

with ϵ_t fractional noise encoding long-range priors. This resonates with predictive processing Hohwy (2013); Friston (2010) and autoregressive cognition models Hinton *et al.* (2006); LeCun *et al.* (2015). Convergence properties connect to evolutionary replicator dynamics Taylor & Jonker (1978); Hofbauer & Sigmund (1998).

E HYDRA Engineering Specification

HYDRA architecture treats RSVP as a multi-head system, each head a distinct (Φ_i, v_i, S_i) module. Persona vectors define boundary data for Φ, shaping functional specialization. Safety is guaranteed by Lyapunov functionals Ashby (1956); Wiener (1948). The architecture addresses AI alignment concerns Bostrom (2014); Russell (2019) by embedding entropic budget constraints into modular decision systems.

F TARTAN Mathematical Framework

TARTAN (Trajectory-Aware Recursive Tiling with Annotated Noise) applies recursive tiling functors to RSVP fields. Each tile carries Gaussian aura fields with metadata. This allows RSVP simulations to maintain coherence across scales. Compositional closure ensures stability of emergent semantic trajectories, echoing synergetic tiling approaches Haken (1983).

G Semantic Merge Operator

Semantic infrastructure requires entropy-respecting composition. For modules $\{M_i\}$:

$$\mathrm{Merge}(\{M_i\}) = \operatorname*{hocolim}_{i \in I} M_i.$$

Obstructions manifest as sheaf cohomology classes Mac Lane (1998); Baez & Stay (2010). The RSVP Emerge functional provides a quantitative constraint:

$$\mathrm{Emerge} = \int_\Omega \lambda_\Phi \|\Phi_1 - \Phi_2\|^2 + \lambda_v \|v_1 - v_2\|^2 + \lambda_S \, \mathrm{KL}(S_1 \| S_2) \, dx.$$

This extends Rissanen's MDL principle Rissanen (1978) and Shannon information Shannon (1948); Cover & Thomas (2006) to semantic gluing.

H EAIMR and Praxicon Formalization

The Effective Assembly Index for Mind Recognition (EAIMR) measures improbability of coherent assembly Levin (2022):

$$\mathrm{EAIMR}(X) = -\log P_{\mathrm{null}}(X).$$

The praxicon, a sheaf of motor/semantic actions in the temporoparietal junction, can be understood as RSVP-structured Deco *et al.* (2008); Oizumi *et al.* (2014); Seth (2016). Failures of v-field divergence link to anosognosia in dementia contexts.

I Empirical Protocols

Three empirical pathways illustrate RSVP:

1. **Cosmology:** Fitting entropy-redshift integrals Jacobson (1995); Verlinde (2011); Padmanabhan (2010); Carney *et al.* (2022) against supernovae and BAO datasets.

2. **Cognition:** RSVP-forced neural PDEs tested with fMRI/EEG Amari (1977); Wilson & Cowan (1972); Deco *et al.* (2008).

3. **Infrastructure:** RSVP MERGE used for adaptive routing in municipal garbage collection or resource distribution Wiener (1948); Ashby (1956).

J Reproducibility and Pipelines

All RSVP simulations must be offline-first and reproducible. Pipelines integrate Bash, Python, and LaTeX with containerization. Deterministic builds align with Landauer's thermodynamic cost of computation Landauer (1961); Bennett (1982). Versioning adopts semantic modules rather than files, consistent with category-theoretic infrastructures.

K Glossary of Neologisms

Lamphron–Lamphrodyne RSVP process of constraint release and entropic smoothing, analogous to Prigogine's dissipative structures Prigogine (1980).

Oblicosm Doctrine Epistemic stance that universes are interpreted through angled perspectives rather than orthogonal absolutes, echoing Ortega's skepticism of mass consensus Ortega y Gasset (1930).

Yarncrawler Infrastructure metaphor of slow, restorative traversal, connecting to ecological computation Levin (2022).

Chain of Memory (CoM) Recursive entropic constraint linking present states to prior budgets, related to Jaynesian entropic inference Jaynes (2003).

Semantic Ladle Mechanism for scooping and redistributing meaning, drawing from cybernetic metaphors Wiener (1948).

References

Alexandrov, M., Kontsevich, M., Schwarz, A., & Zaboronsky, O. 1997. The Geometry of the Master Equation and Topological Quantum Field Theory. *Int. J. Mod. Phys. A*, **12**, 1405–1429.

Amari, Shun-ichi. 1977. Dynamics of Pattern Formation in Lateral-Inhibition Type Neural Fields. *Biol. Cybern.*, **27**, 77–87.

Ashby, W. Ross. 1956. *An Introduction to Cybernetics.* Chapman & Hall.

Baez, John C., & Stay, Mike. 2010. *Physics, Topology, Logic and Computation: A Rosetta Stone.* Lecture Notes in Physics, vol. 813. Springer.

Barrow, John D., & Magueijo, Joao. 2020. Temperature and Entropy of Cosmological Horizons. *Phys. Rev. D*, **101**(4), 043521.

Batalin, I. A., & Vilkovisky, G. A. 1981. Gauge Algebra and Quantization. *Physics Letters B*, **102**, 27–31.

Bennett, Charles H. 1982. The Thermodynamics of Computation—A Review. *International Journal of Theoretical Physics*, **21**(12), 905–940.

Bostrom, Nick. 2014. *Superintelligence: Paths, Dangers, Strategies.* Oxford University Press.

Carney, D., Cèspedes, S., *et al.* 2022. Entanglement, Gravity, and Thermodynamics: A Review. *Rev. Mod. Phys.*, **94**(1), 015001.

Caticha, Ariel. 2012. *Entropic Inference and the Foundations of Physics.* University at Albany.

Cattaneo, Alberto S., & Felder, Giovanni. 2000. A Path Integral Approach to the Kontsevich Quantization Formula. *Commun. Math. Phys.*, **212**, 591–611.

Cover, Thomas M., & Thomas, Joy A. 2006. *Elements of Information Theory.* 2nd edn. Wiley.

Deco, Gustavo, Jirsa, Viktor K., Robinson, Peter A., Breakspear, Michael, & Friston, Karl. 2008. The Dynamic Brain: From Spiking Neurons to Neural Masses and Cortical Fields. *PLoS Comput. Biol.*, **4**, e1000092.

Dennett, Daniel C. 1991. *Consciousness Explained.* Little, Brown and Co.

Friston, Karl. 2010. The Free-Energy Principle: A Unified Brain Theory? *Nat. Rev. Neurosci.*, **11**, 127–138.

Haken, Hermann. 1983. *Synergetics: An Introduction.* Springer.

Hinton, Geoffrey E., Osindero, Simon, & Teh, Yee-Whye. 2006. A Fast Learning Algorithm for Deep Belief Nets. *Neural Computation*, **18**(7), 1527–1554.

Hofbauer, Josef, & Sigmund, Karl. 1998. *Evolutionary Games and Population Dynamics*. Cambridge University Press.

Hohwy, Jakob. 2013. *The Predictive Mind*. Oxford University Press.

Jacobson, Ted. 1995. Thermodynamics of Spacetime: The Einstein Equation of State. *Phys. Rev. Lett.*, **75**, 1260–1263.

Jaynes, E. T. 2003. *Probability Theory: The Logic of Science*. Cambridge University Press.

Landauer, R. 1961. Irreversibility and Heat Generation in the Computing Process. *IBM Journal of Research and Development*, **5**(3), 183–191.

LeCun, Yann, Bengio, Yoshua, & Hinton, Geoffrey. 2015. Deep Learning. *Nature*, **521**, 436–444.

Lehalleur, Simon Pepin, Hoogland, Jesse, Farrugia-Roberts, Matthew, Wei, Susan, Oldenziel, Alexander Gietelink, Wang, George, Carroll, Liam, & Murfet, Daniel. 2025. You Are What You Eat - AI Alignment Requires Understanding How Data Shapes Structure and Generalisation. *arXiv preprint arXiv:2502.04810*, February.

Levin, Michael. 2022. Technological Approach to Mind Everywhere: An Experimentally-Grounded Framework for Understanding Diverse Bodies and Minds. *Frontiers in Systems Neuroscience*, **16**, 768201.

Mac Lane, Saunders. 1998. *Categories for the Working Mathematician*. 2nd edn. Graduate Texts in Mathematics, vol. 5. Springer.

Nowak, Martin A. 2006. *Evolutionary Dynamics: Exploring the Equations of Life*. Harvard University Press.

Oizumi, Masafumi, Albantakis, Larissa, & Tononi, Giulio. 2014. From the Phenomenology to the Mechanisms of Consciousness: Integrated Information Theory 3.0. *PLOS Computational Biology*, **10**(5), e1003588.

Ortega y Gasset, José. 1930. *The Revolt of the Masses*. W. W. Norton & Company.

Padmanabhan, T. 2010. Thermodynamical Aspects of Gravity: New Insights. *Rept. Prog. Phys.*, **73**, 046901.

Pantev, Tony, Toën, Bertrand, Vaquié, Michel, & Vezzosi, Gabriele. 2013. Shifted Symplectic Structures. *Publ. Math. IHES*, **117**, 271–328.

Parr, Thomas, Pezzulo, Giovanni, & Friston, Karl J. 2022. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press.

Prigogine, Ilya. 1980. *From Being to Becoming: Time and Complexity in the Physical Sciences*. W. H. Freeman.

Rissanen, Jorma. 1978. Modeling by Shortest Data Description. *Automatica*, **14**(5), 465–471.

Russell, Stuart. 2019. *Human Compatible: Artificial Intelligence and the Problem of Control*. Viking.

Seth, Anil K. 2016. Theories and Measures of Consciousness: An Extended Framework. *Proceedings of the National Academy of Sciences*, **113**(17), 4903–4910.

Shannon, Claude E. 1948. A Mathematical Theory of Communication. *Bell System Technical Journal*, **27**(3), 379–423.

Smolin, Lee. 2019. *Einstein's Unfinished Revolution: The Search for What Lies Beyond the Quantum*. Penguin.

Taylor, Peter D., & Jonker, Leo B. 1978. Evolutionarily Stable Strategies and Game Dynamics. *Mathematical Biosciences*, **40**(1–2), 145–156.

Tononi, Giulio. 2012. *Phi: A Voyage from the Brain to the Soul*. Pantheon.

Verlinde, Erik P. 2011. On the Origin of Gravity and the Laws of Newton. *JHEP*, **04**, 029.

Wiener, Norbert. 1948. *Cybernetics: Or Control and Communication in the Animal and the Machine*. MIT Press.

Wilson, Hugh R., & Cowan, Jack D. 1972. Excitatory and Inhibitory Interactions in Localized Populations of Model Neurons. *Biophys. J.*, **12**, 1–24.

Yudkowsky, Eliezer, & Soares, Nate. 2025. *If Anyone Builds It, Everyone Dies: Why Superhuman AI Would Kill Us All*. New York, NY: Little, Brown and Company.